Agile Development Process and Devops

(ISWE406L)

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64. Customer Relationship Management (CRM) System

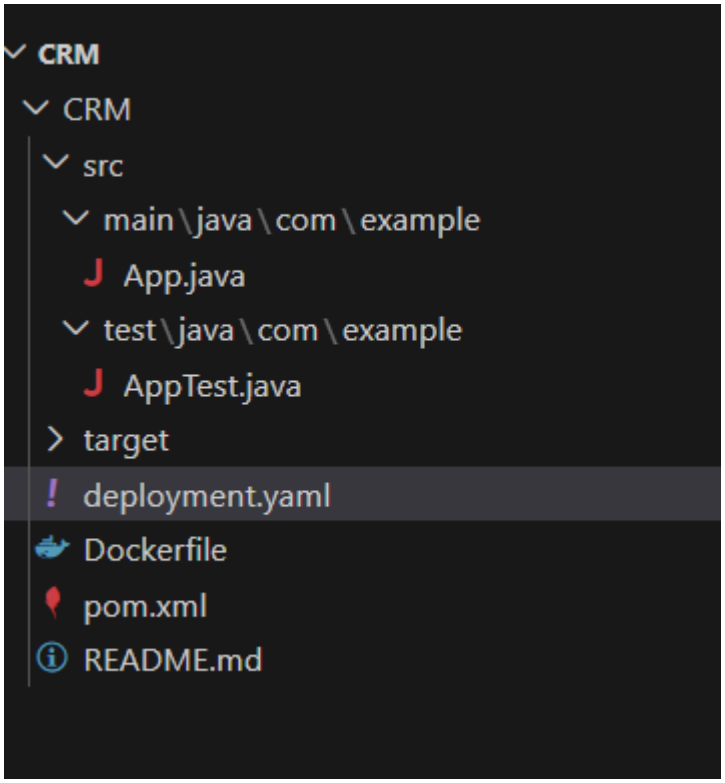
Problem Statement:
Ensure customer data and interactions are handled correctly before deployment.

Tasks:

- Develop CRM data handling logic
- Write JUnit tests for:
  - Data validation
  - Duplicate customer entries
- CI/CD pipeline for automated deployment
- Deploy on Kubernetes

Real-world relevance: Sales and customer management tools

Folder Structure:


CRM
CRM
src
main\java\com\example
App.java
test\java\com\example
AppTest.java
target
deployment.yaml
Dockerfile
pom.xml
README.md

App.java:

```java
import java.util.*;

public class App {

    static class Customer {
        String email;
        String name;

        Customer(String name, String email) {
            this.name = name;
            this.email = email;
        }
    }

    static List<Customer> customers = new ArrayList<>();

    // ADD CUSTOMER (with validation & duplicate check)
    public static boolean addCustomer(String name, String email) {

        if (name == null || name.isEmpty() || email == null || !email.contains("@")) {
            return false; // invalid data
        }

        for (Customer c : customers) {
            if (c.email.equals(email)) {
                return false; // duplicate customer
            }
        }

        customers.add(new Customer(name, email));
        return true;
    }

    public static int totalCustomers() {
        return customers.size();
    }

```

```java
    public static void main(String[] args) {
        addCustomer("Asha", "asha@gmail.com");
        addCustomer("John", "john@gmail.com");

        System.out.println("Total Customers: " + totalCustomers());
    }
}
```

AppTest.java:

```java
import org.junit.Test;
import static org.junit.Assert.*;

public class AppTest {

    @Test
    public void testValidCustomer() {
        App.customers.clear();
        boolean result = App.addCustomer("Asha", "asha@gmail.com");
        assertTrue(result);
    }

    @Test
    public void testInvalidEmail() {
        App.customers.clear();
        boolean result = App.addCustomer("Asha", "ashagmail.com");
        assertFalse(result);
    }

    @Test
    public void testDuplicateCustomer() {
        App.customers.clear();
        App.addCustomer("Asha", "asha@gmail.com");
        boolean result = App.addCustomer("Asha2", "asha@gmail.com");
        assertFalse(result);
    }
}
```

pom.xml:

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>

  <groupId>crm</groupId>
  <artifactId>crm-system</artifactId>
  <version>1.0</version>

  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

Dockerfile:

pom.xml Dockerfile

CRM > Dockerfile

```
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY target/*.jar app.jar
CMD ["java", "-jar", "app.jar"]
```

Local testing using Maven:

```
PS C:\Users\gopik\Downloads\CRM> cd CRM
PS C:\Users\gopik\Downloads\CRM\CRM> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------------< crm:crm-system >-------------------------
[INFO] Building crm-system 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-c
)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/35/maven-plugins-
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/35/maven-plugins-3
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/35/maven-parent-35.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-parent/35/maven-parent-35.pom (45
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/apache/25/apache-25.pom
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/apache/25/apache-25.pom (21 kB at 360 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-clean-plugin/3.2.0/maven-c
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/ma
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-resources-plugin/3.4.0/mav
127 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/45/maven-plugins-
```

```
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.123 s -- in AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  17.704 s
[INFO] Finished at: 2026-04-12T19:14:02+05:30
```

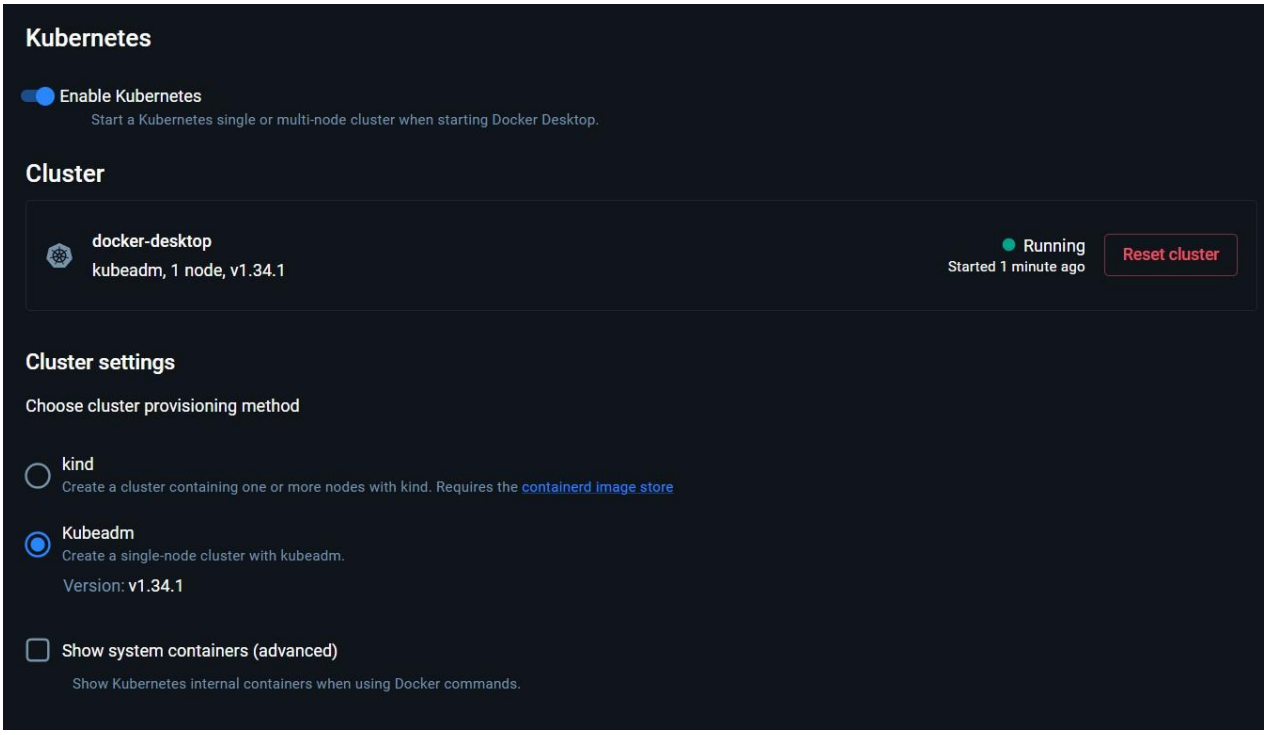
Pushed to github:

```
PS C:\Users\gopik\Downloads\CRM> git init
Initialized empty Git repository in C:/Users/gopik/Downloads/CRM/.git/
PS C:\Users\gopik\Downloads\CRM> git clone
fatal: You must specify a repository to clone.
```

```
Started by user Gopikasathish
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\ABC
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Tool Install)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout Git)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/gopika-sathish/CRM.git
 > git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\ABC # timeout=10
Fetching upstream changes from https://github.com/gopika-sathish/CRM.git
```

```
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running AppTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.652 s -- in
AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  01:12 min
[INFO] Finished at: 2026-04-12T19:45:19+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Docker Engine and kubernetes engine started on Docker Desktop:



MVN CLEAN PACKAGE:

```
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\gopik\Downloads\CRM\CRM> docker logs crm-container
no main manifest attribute, in crm-system-1.0.jar
PS C:\Users\gopik\Downloads\CRM\CRM> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:order-tracking >--------------------
[INFO] Building order-tracking 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.11
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.11
 kB at 12 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/39/maven-plu
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-plugins/39/maven-plug
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.11
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-compiler-plugin/3.11
kB at 876 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.3.0/mav
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/plugins/maven-jar-plugin/3.3.0/mave
B/s)
```

```
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.26/plexus-interpolation-1.26.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/maven/maven-archiver/3.6.0/maven-archiver-3.6.0.jar (26 kB at 197 kB/s)
Downloading from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.4.2/plexus-utils-3.4.2.jar
Downloaded from central: https://repo.maven.apache.org/maven2/org/tukaani/xz/1.9/xz-1.9.jar (116 kB at 678 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-archiver/4.4.0/plexus-archiver-4.4.0.jar (211 kB at 1.2 MB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-interpolation/1.26/plexus-interpolation-1.26.jar (85 kB at 445 k
B/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/codehaus/plexus/plexus-utils/3.4.2/plexus-utils-3.4.2.jar (267 kB at 851 kB/s)
Downloaded from central: https://repo.maven.apache.org/maven2/org/apache/commons/commons-compress/1.21/commons-compress-1.21.jar (1.0 MB at 3.2 MB/s)
[INFO] Building jar: C:\Users\gopik\Downloads\CRM\CRM\target\order-tracking-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  11.847 s
[INFO] Finished at: 2026-04-12T23:10:41+05:30
[INFO] ------------------------------------------------------------------------
PS C:\Users\gopik\Downloads\CRM\CRM>
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\gopik\Downloads\CRM> docker tag crm-app gopikasathish/crm-app:latest
PS C:\Users\gopik\Downloads\CRM> docker push gopikasathish/crm-app:latest
The push refers to repository [docker.io/gopikasathish/crm-app]
5fc84f25fa54: Layer already exists
b35d291c75ff: Already exists
336e5a704d56: Layer already exists
a149c623348d: Layer already exists
a1e6e1a167f1: Layer already exists
f00bd8c488bb: Layer already exists
a7428f4f5c1d: Layer already exists
689b91d88a0f: Pushed
latest: digest: sha256:663730b322cdb9aa97aa1d1d1557739c2789c7ded775f6ba15d8e8da369de40a size: 856
PS C:\Users\gopik\Downloads\CRM>
```

```
PS C:\Users\gopik\Downloads\CRM> cd CRM
PS C:\Users\gopik\Downloads\CRM\CRM> kubectl apply -f deployment.yaml
deployment.apps/crm-deployment created
PS C:\Users\gopik\Downloads\CRM\CRM>
```